DSR301M · PRESENT 3

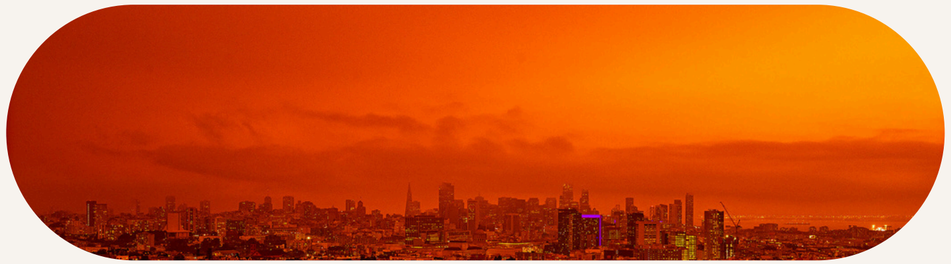
PBL: COMPARATIVE US AQI TIME SERIES FORECASTING IN CALIFORNIA

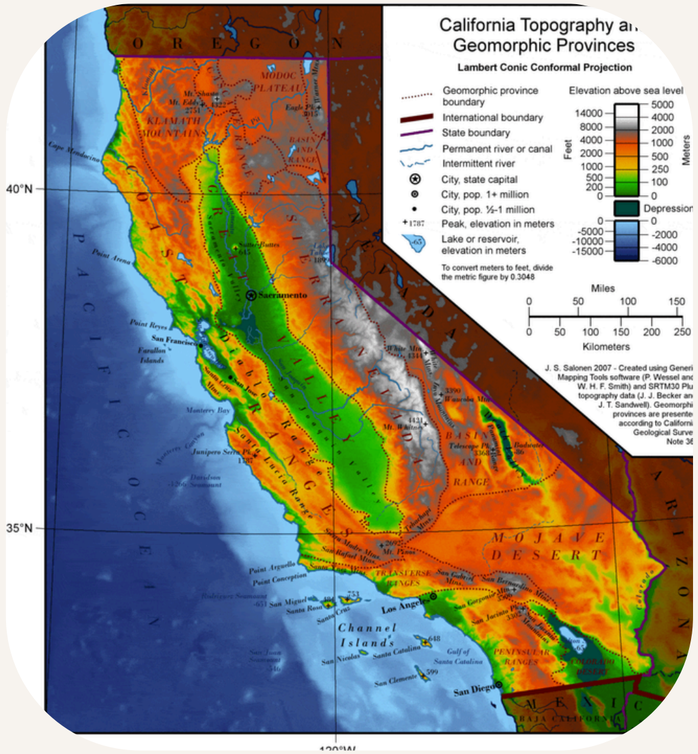
WITH EXTREME-EVENT EVALUATION

GROUP 5

Wildfire Crisis & Hazardous Smoke Hazards in California

Historic August Complex 2020 Megafire & Acute Respiratory Shocks





HISTORIC AUGUST COMPLEX PEAK: 496.0 AQI

- Surpasses Hazardous threshold (301+) by +195 pts
- San Jose (Sept 2020) · Daylight blotted out across Bay Area

ACUTE PUBLIC HEALTH BURDEN

- >90% surge in emergency hospital visits
- Wildfire PM2.5 exceeded WHO guidelines by >10x

SEVERE DATA IMBALANCE

- >85% of days are clean air baseline (AQI ≤ 50)
- Standard models miss rare disaster spikes

Dual Operational Horizons & Extreme Wildfire Upgrade

Designing Purpose-Built Forecasting Architectures for Public Health & Disaster Action

HORIZON 1 · NOWCASTING 1h (t+1)

- Action: Immediate worker alerts & dynamic traffic routing
- Driver: Sensor inertia (lags 0–3h, rolling stats, 1h diff)
- SOTA Model: Spatio-Temporal LSTM (MAE 2.76, R^2 0.962)

HORIZON 2 · DAY-AHEAD 24h (t+24)

- Action: School closures, evacuation & ICU hospital staging
- Driver: NWP future weather (Wt+24), diurnal cycles & wind vector
- SOTA Model: Voting Ensemble (MAE 11.30, R^2 0.617)

MILESTONE 3 · WILDFIRE REGIME UPGRADE

- Gating Trigger: Santa Ana alignment + VPD > 1.8 kPa

- 1h Nowcasting: Peak error: 64.9 → 45.8 AQI (MAE -13.1%)
- 24h Day-Ahead: Recall (>150): 5.3% → 20.1% (3.8x jump, F1 3.1x)

- Disaster Peak: Recovers +32.8 AQI on Palisades Fire (150.9 → 183.8)

Benchmark Model Suite: 11 Models Across 6 Algorithm Families

BASELINES
- Persistence (y_t)
- Climatology (Hist. mean)
- Zero-training benchmark

LINEAR L2
- Ridge Regression
- Analytical closed-form
- Fit time: ~0.18s

BAGGING
- Random Forest
- 150 Decision Trees
- Subsampling variance cut

GBDT SOTA
- LightGBM (leaf-wise)
- XGBoost (exact Hessian)
- CatBoost (oblivious)

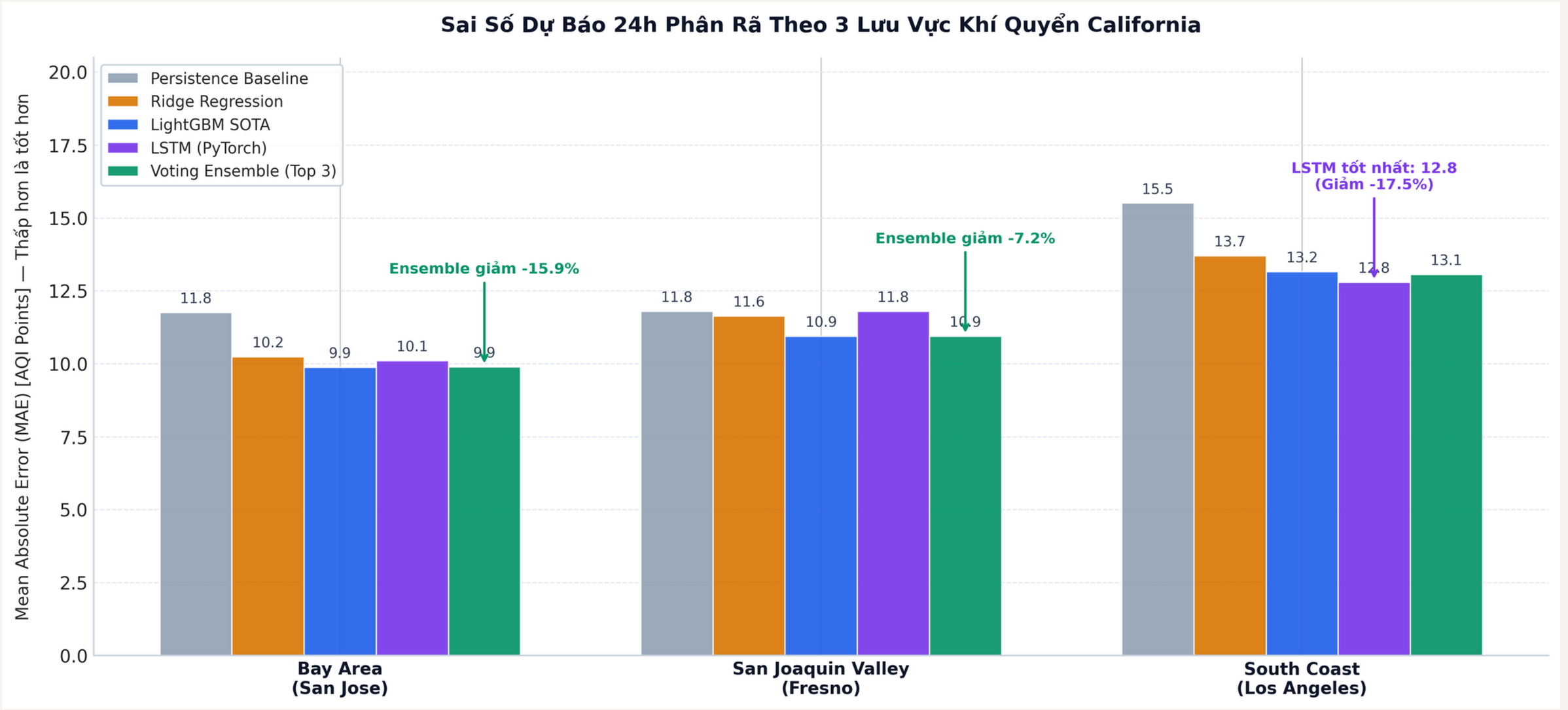
DEEP LEARNING
- Tabular MLP Regressor (64, 32 units)
- [NEW] Spatio-Temporal LSTM (PyTorch):
  - 2-Layer LSTM + 8-dim Station Embedding
  - Smooth L1 (Huber) Loss
  - ★ Top 1 at 1h (MAE 2.76, R^2 0.962)

TOP-3 GBDT VOTING ENSEMBLE: Blending LightGBM + XGBoost + CatBoost (★ Top 1 at 24h: MAE 11.30)

[NEW] WILDFIRE SPECIALIST (2-STAGE GBDT): Regime-Switching Gating (★ 3.8x Recall on >150 AQI)

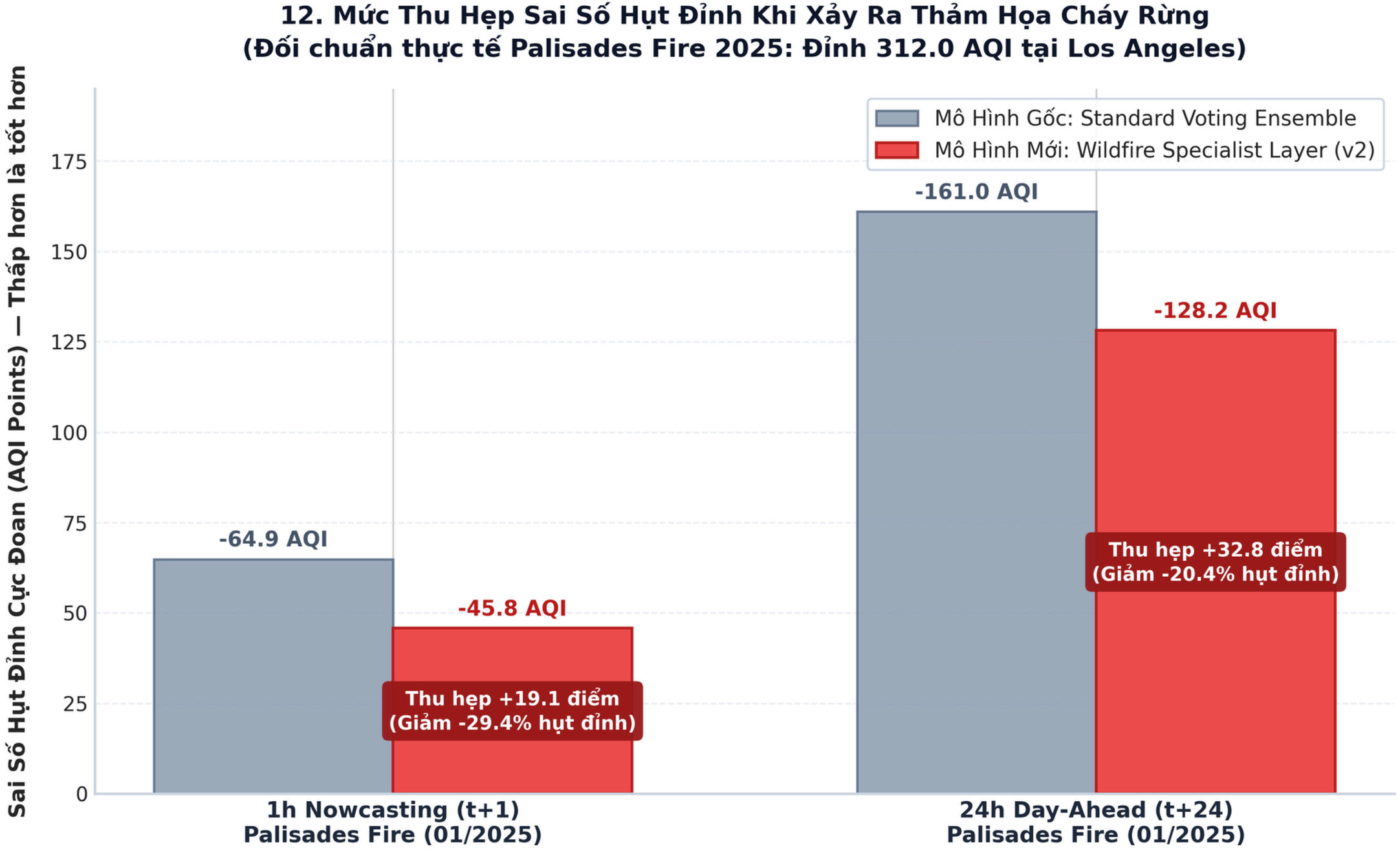
OFFICIAL COMPREHENSIVE LEADERBOARD

Model	MAE 1h	R^2 1h	MAE 24h	R^2 24h	Fit (24h)
Climatology	17,53	0,150	17,45	0,155	2,37
Persistence	3,61	0,948	13,02	0,460	0,00
Ridge Reg.	3,72	0,948	11,86	0,586	0,24
Random Forest	3,33	0,954	11,53	0,604	8,93
LightGBM	3,13	0,954	11,33	0,617	0,41
XGBoost	3,10	0,954	11,36	0,616	0,67
CatBoost	3,13	0,954	11,40	0,608	1,45
Tabular MLP	3,26	0,958	13,08	0,501	73,10
LSTM (PyTorch)	**2,76**	**0,962**	11,56	0,584	121,12
Voting Ensem.	3,06	0,956	**11,30**	**0,617**	0,20
Wildfire Specialist	3,05	0,958	11,46	0,612	0,30

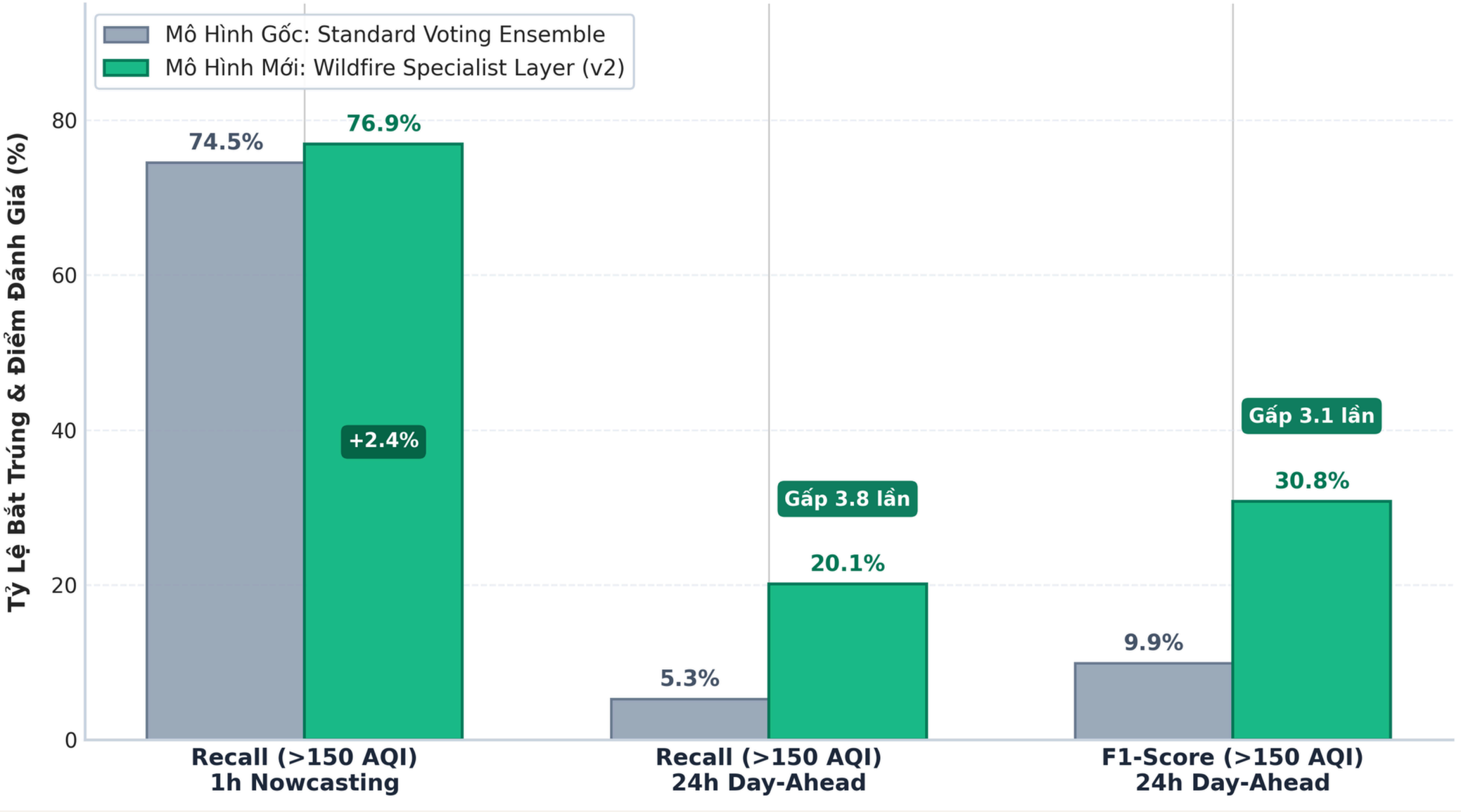

Sai Số Dự Báo 24h Phân Rã Theo 3 Lưu Vực Khí Quyển California
Mean Absolute Error (MAE) [AQI Points] — Thấp hơn là tốt hơn
Persistence Baseline
Ridge Regression
LightGBM SOTA
LSTM (PyTorch)
Voting Ensemble (Top 3)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
11.8
10.2
9.9
10.1
9.9
Ensemble giảm -15.9%
11.8
11.6
10.9
11.8
10.9
Ensemble giảm -7.2%
15.5
13.7
13.2
12.8
13.1
LSTM tốt nhất: 12.8
(Giảm -17.5%)
Bay Area
(San Jose)
San Joaquin Valley
(Fresno)
South Coast
(Los Angeles)

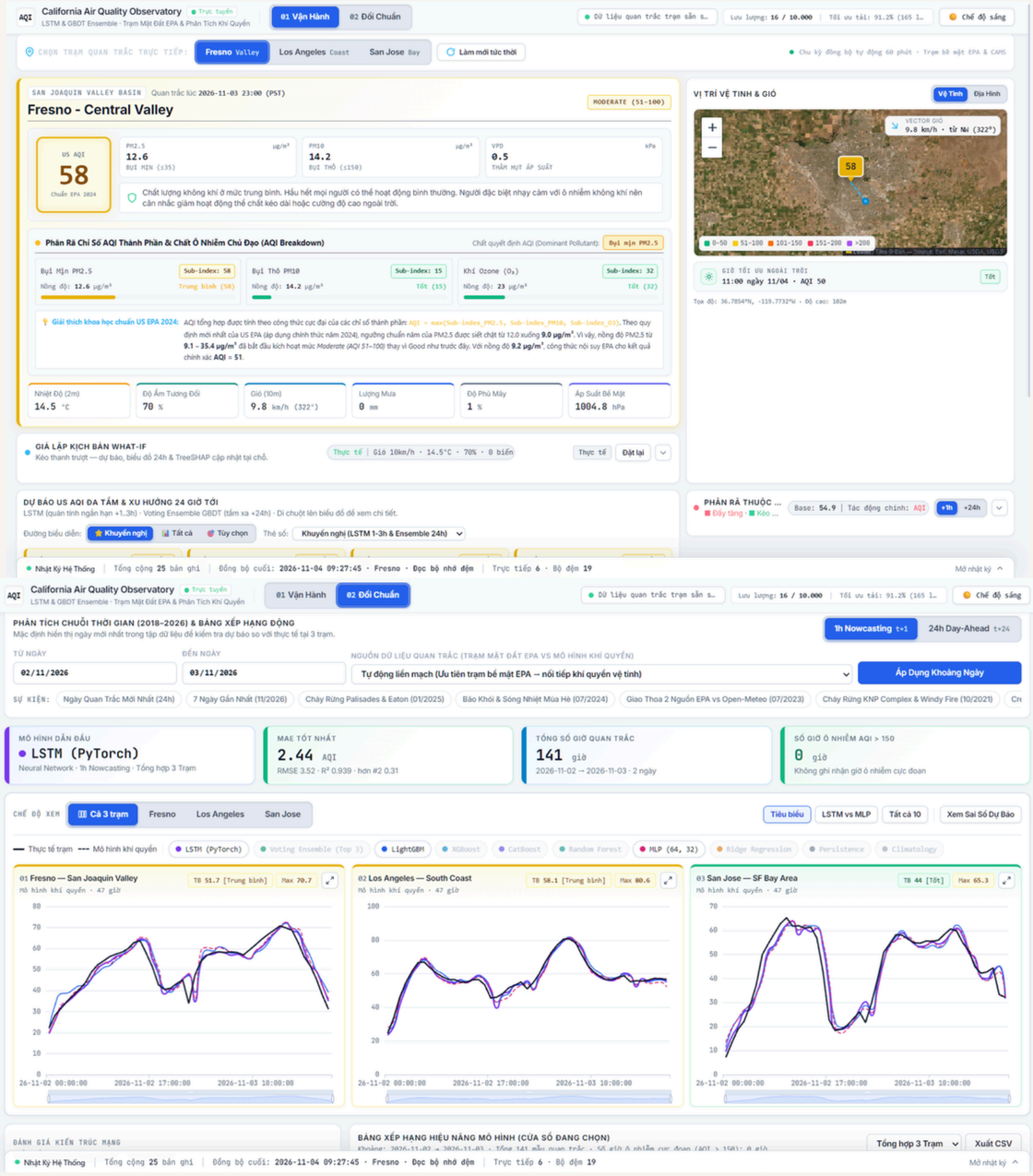
Wildfire Extreme Shock Resilience: 24h Day-Ahead Leaderboard

Model	MAE Ngày Thường	MAE Bão Khói	Recall Bắt Trúng (%)	F1-Score
Climatology	14,92	102,06	0	0
LSTM (PyTorch)	9,94	71,04	0	0
Tabular MLP	11,69	68,11	10,34	0,179
CatBoost	9,97	64,95	4,08	0,077
Voting Ensemble	9,91	62,64	5,33	0,099
XGBoost	9,98	62,49	8,15	0,149
LightGBM	9,96	60,81	6,9	0,123
Ridge Reg.	10,62	58,92	6,58	0,118
Random Forest	10,25	58,46	15,05	0,247
Wildfire Specialist	10,27	52,88	20,06	0,308


12. Mức Thu Hẹp Sai Số Hụt Đỉnh Khi Xảy Ra Thảm Họa Cháy Rừng
(Đối chuẩn thực tế Palisades Fire 2025: Đỉnh 312.0 AQI tại Los Angeles)
Sai Số Hụt Đỉnh Cực Đoan (AQI Points) — Thấp hơn là tốt hơn
Mô Hình Gốc: Standard Voting Ensemble
Mô Hình Mới: Wildfire Specialist Layer (v2)
0
25
50
75
100
125
150
175
-64.9 AQI
-45.8 AQI
Thu hẹp +19.1 điểm
(Giảm -29.4% hụt đỉnh)
-161.0 AQI
-128.2 AQI
Thu hẹp +32.8 điểm
(Giảm -20.4% hụt đỉnh)
1h Nowcasting (t+1)
Palisades Fire (01/2025)
24h Day-Ahead (t+24)
Palisades Fire (01/2025)

13. Bước Nhảy Vọt Năng Lực Nhận Diện Cảnh Báo Sớm Giờ Nguy Hại (AQI > 150)
(Đánh giá Out-of-Sample trên toàn bộ 26.207 giờ kiểm định năm 2025)





LIVE DEMO
Web Dashboard